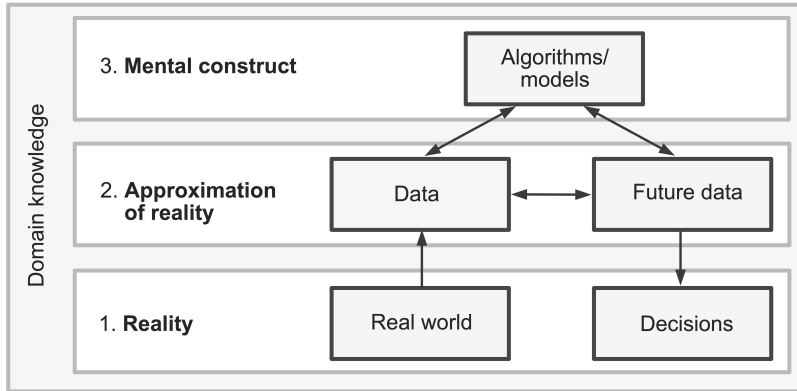


Model diagnostics

Ciaran Evans

Model assessment

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



Question: Is our model useful?

Motivating example: Dengue data

Data: Data on 5720 Vietnamese children, admitted to the hospital with possible dengue fever. Variables include:

- ▶ *Sex*: patient's sex (female or male)
- ▶ *Age*: patient's age (in years)
- ▶ *WBC*: white blood cell count
- ▶ *PLT*: platelet count
- ▶ other diagnostic variables. . .
- ▶ *Dengue*: whether the patient has dengue (0 = no, 1 = yes)

Motivating example: Dengue data

Y_i = dengue status (0 = negative, 1 = positive)

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 WBC_i$$

Assumptions and diagnostics

What assumptions do we make in a *linear* regression model?

- shape / linearity $\leftarrow$ scatterplot of y vs. x ,
residuals $(y - \hat{y})$ against x or $\hat{y}$
- Don't need these for logistic regression
- zero mean, normal errors (ϵ_i)
 - constant variance (of ϵ_i)
 - independence
 - randomness
- } Think about data generating process

Logistic regression assumptions

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Age}_i$$

Shape: log-odds are a linear function of the explanatory variables

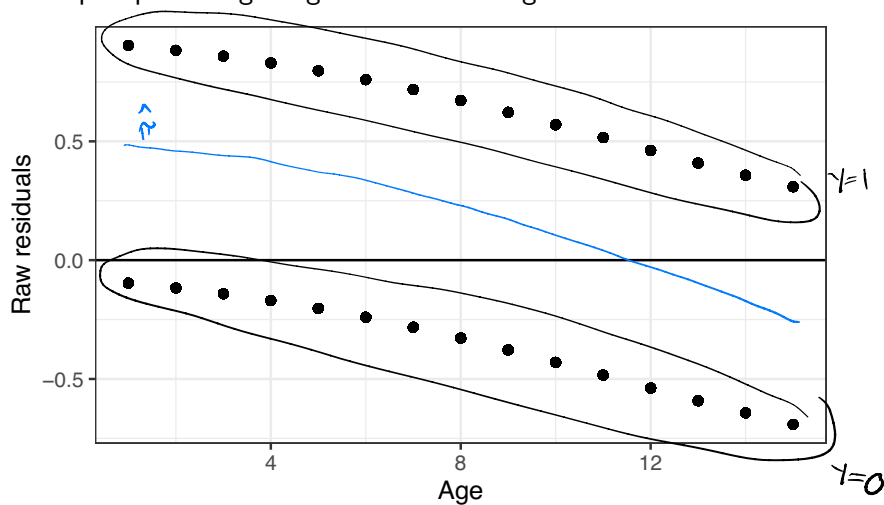
Independence: observations are independent (need this for ML!))

Randomness: data come from a random process

Don't use raw residuals for logistic regression

Raw residuals: $Y_i - \hat{\pi}_i \rightarrow$ either $1 - \hat{\pi}_i$
or $0 - \hat{\pi}_i$

Example: predicting dengue status with Age



Alternative: empirical logits

(based on observed data, not on a fitted model)

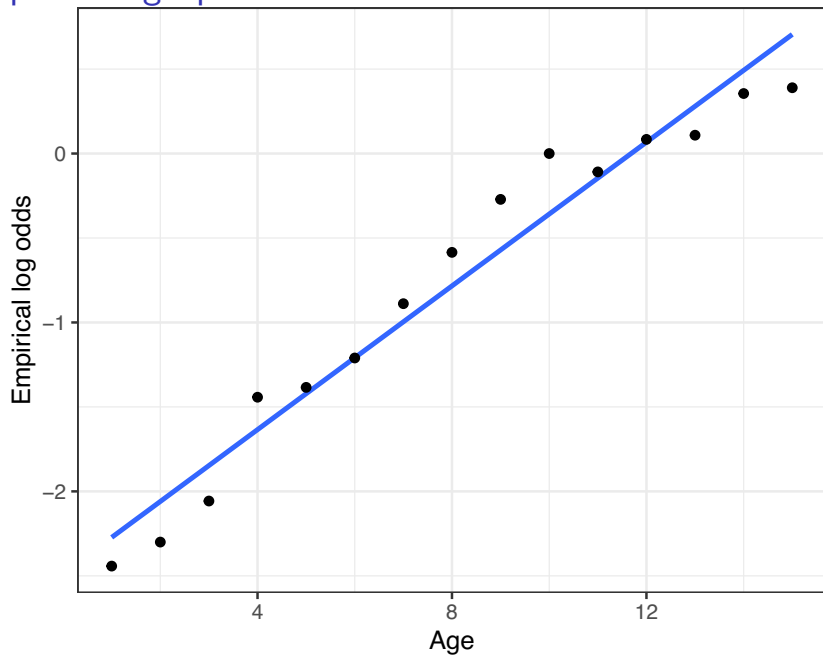
For each Age, calculate the *empirical* log odds of dengue (aka *empirical logits*):

```
## # A tibble: 15 x 5
```

##	Age	NoDengue	YesDengue	p	log_odds
##	<int>	<int>	<int>	<dbl>	<dbl>
##	1	1	184	16	0.08 -2.44
##	2	2	419	42	0.0911 -2.30
##	3	3	532	68	0.113 -2.06
##	4	4	457	108	0.191 -1.44
##	5	5	487	122	0.200 -1.38
##	6	6	369	110	0.230 -1.21
##	7	7	360	148	0.291 -0.889
##	8	8	289	161	0.358 -0.585
##	9	9	244	186	0.433 -0.271
##	10	10	185	185	0.5 0
##	11	11	165	148	0.473 -0.109
##	12	12	126	137	0.521 0.0837
##	13	13	96	107	0.527 0.108

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right)$$

Empirical logit plot



Empirical logits with a continuous variable

Trying with WBC:

```
## # A tibble: 6 x 5
```

```
##      WBC NoDengue YesDengue      p log_odds
##    <dbl>   <int>    <int> <dbl>   <dbl>
## 1  0.9      0        1      1     Inf
## 2  1.15     0        2      1     Inf
## 3  1.23     0        1      1     Inf
## 4  1.25     0        1      1     Inf
## 5  1.54     0        3      1     Inf
## 6  1.58     0        1      1     Inf
```

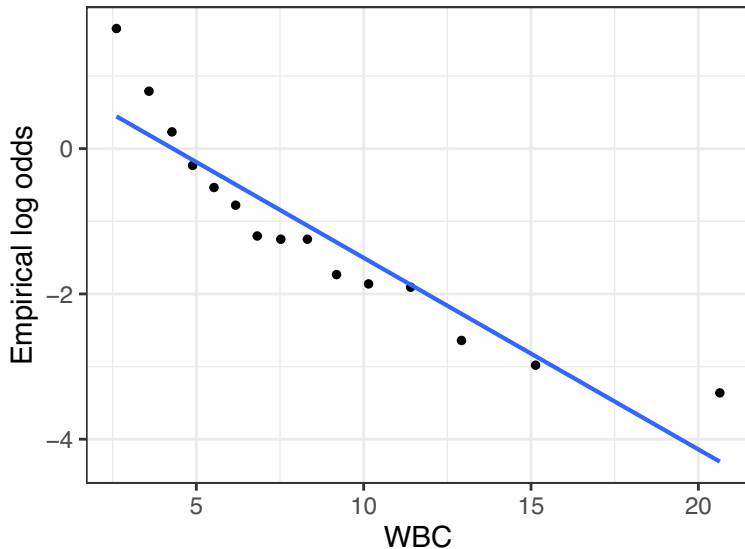
$$\log\left(\frac{1}{0}\right) = \log(\infty) = \infty ?$$

What has gone wrong? What should we do differently?

WBC is continuous $\rightarrow$ only see 1 or 2 people w/ each value

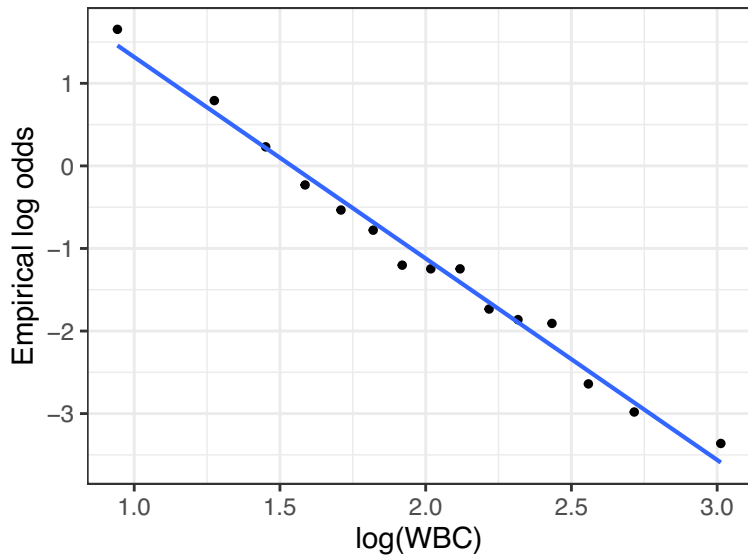
Bin (group) observations by WBC, calculate empirical log odds for each bin
e.g. 0.5 - 2, 2 - 3.5, 3.5 - 5, etc.

Binned empirical logit plots

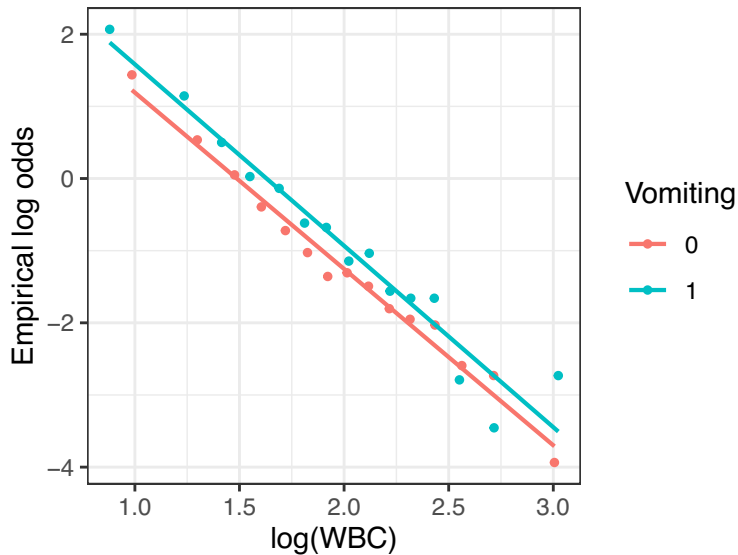


Does the shape assumption seem reasonable?

Binned empirical logit plots



Binned empirical logit plots



Limitations of empirical logit plots

- ▶ Does not assess a fitted model (performed *before* fitting the model)
- ▶ One point per *bin*, not one point per *observation*
- ▶ Can only plot one or two explanatory variables at a time

Class activity

- ▶ Work on the class activity in R (linked on Canvas and the course website)
- ▶ You are welcome to work in groups
- ▶ At the end of class, submit HTML file on Canvas (one per group, listing all group member names)